

Yanbo Zhou

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About me

I am currently pursuing my Ph.D. at UCSD. I have broad interests in storage and memory systems, including cloud storage, flash storage, persistent memory, CXL, and software-hardware co-design. Over the past years, I have contributed many innovations in cloud storage from concept to completion. Most of them are currently deployed and have benefited various block storage products of Alibaba Cloud.

Education

2023 - now	University of California, San Diego Ph.D. in Computer Science Advisor: Dr. Steven Swanson	San Diego, CA, USA
2016 - 2019	East China Normal University M.Eng. in Software Engineering	Shanghai, China
2012 - 2016	Taiyuan University of Technology B.Eng. in Software Engineering	Taiyuan, China

Professional Experience

2019 - 2023	Alibaba Cloud Senior Software Engineer , Storage Group Worked on storage software pathfinding activities for block storage products of Alibaba Cloud and had contributed to the following storage products: 1. I2/I2ne Instance: The second-gen high-performance and low-latency local storage. This is a software-optimized architecture towards NVMe TLC SSDs. [link] 2. D3c Instance: The third-gen cost-effective and high-capacity local storage. This is the first time enabling NVMe QLC SSDs and Zoned Namespace (ZNS) technology in Alibaba Cloud. [link] 3. I4 Instance: The forth-gen high-performance and low-latency local storage. This is a software/hardware co-designed architecture optimized for NVMe TLC SSDs and bare-metal. 4. ESSD-PLX: The fastest elastic block storage volume (300us end-to-end latency) of Alibaba Cloud based on Optane Persistent Memory cache and RDMA technology. [link]	Hangzhou, China
2017 - 2018	Intel Corporation Development Intern , Data Center Group Worked on open-source storage project SPDK (storage performance development kit [link]) and had contributed to the following project: 1. Vhost-NVMe: First full-stack userspace NVMe virtualization solution. This technology is used as standard storage virtualization protocol for multiple storage products in Alibaba Cloud EBS afterwards, e.g., EBS ESSD, Local Storage Instance D3c. [link]	Shanghai, China

Publications

Book Chapters

2020	Linux Open Source Storage: from Ceph to Container (Chinese Edition). In the Publishing House of Electronics Industry, 2019. [link]
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Conference Proceedings

- 2024 **Sonic: the Next-Gen Local Disks for the Cloud.**
Yanbo Zhou, Erci Xu, Li Zhang, Kapil Karkra, Mariusz Barczak, Wayne Gao, Wojciech Malikowski, Mateusz Kozłowski, Łukasz Łasek, Ruiming Lu, Feng Yang, Lilong Huang, Xiaolu Zhang, Wenrui Li, Jinhui Li, Keqiang Niu, Jiaji Zhu, Jiesheng Wu.
To appear in EuroSys 2024.
- 2020 **Spool: Reliable Virtualized NVMe Storage Pool in Public Cloud Infrastructure.**
Shuai Xue, Shang Zhao, Quan Chen, Gang Deng, Zheng Liu, Jie Zhang, Zhuo Song, Tao Ma, Yong Yang, Yanbo Zhou, Keqiang Niu, Sijie Sun, Minyi Guo.
In the Proceedings of the USENIX Annual Technical Conference (ATC), 2020.
- 2018 **Write-aware Data Allocation on Heterogeneous Memory Architecture with Minimum Cost.**
Yanbo Zhou, Shouzheng Gu, Lixia Zheng, Edwin H.-M. Sha, Qingfeng Zhuge.
In the Proceedings of the IEEE International Conference on Embedded and Real-Time Computing System and Applications (RTCSA), 2018.
- 2018 **Accelerating I/Os in virtual machines on physical NVMe SSDs via user space vhost target.**
Ziye Yang, Changpeng Liu, Yanbo Zhou, Xiaodong Liu, Gang Cao.
In the Proceedings of the IEEE International Symposium on Cloud and Services Computing (SC2), 2018.
- Others**
- 2023 **A Media-Aware Cloud Storage Acceleration Layer (CSAL) Cache Solution with Intel Optane SSDs for Alibaba ECS Local Disk D3C Service.**
Yanbo Zhou, Li Zhang, Kapil Karkra, Wayne Gao, Chunhong Mao, Mariusz Barczak.
In Intel White Paper, 2023. [\[link\]](#)

Talks

- 2023 **Best SPDK Practices: Lessons from Five Years of Storage Evolution in Alibaba Cloud,** Yanbo Zhou.
In SPDK PRC Virtual Forum, 2023. [\[link\]](#)
- 2022 **CSAL: Enabling Unprecedented Performance and Capacity Values with Optane and QLC Flash,** Yanbo Zhou, Kapil Karkra, Qinghua Ye, Li Zhang, Mariusz Barczak, Wojciech Malikowski, Wayne Gao, Greg Scott, Ron Thornburg.
In SNIA Storage Developer Conference, Silicon Valley, CA, 2022. [\[link\]](#)
- 2022 **O+Q in the cloud: Removing QLC Write-Amplification through Intel Optane SSD with SPDK WSR,** Yanbo Zhou.
In SPDK PRC Virtual Forum, 2022. [\[link\]](#)

Professional Skills

Programming Languages

- C/C++ (Proficient)
- Python, Shell (Familiar)
- Linux kernel and userspace programming skills

Systems

- Linux Kernel: filesystem, block, and driver layer
- Virtualization Software: QEMU, Vhost-User
- Protocol: NVMe, NVMeOF, Virtio

Updated September 2023